The Dual-Layer Economy

A Macroeconomic Architecture for Resilience (v2.0)

Working Paper

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Abstract

This paper presents the Dual-Layer Economy (DLE), a stock-flow consistent hybrid macroeconomic architecture designed to achieve bounded dynamics and shock recovery under structural financial instability. Building on prior work establishing architectural separation (Rentschler, 2025a) and transition dynamics (Rentschler, 2025b), this paper introduces Resilience Capital (Z), an endogenous state variable that accumulates through targeted investment and modifies both effective productivity and depreciation, enabling recovery from shocks that would otherwise produce persistent low-output traps.

The model is implemented as a projected hybrid dynamical system with regime switching, automatic fiscal stabilizers, and guard-based state constraints. Analytical examination of a reduced smooth-core subsystem shows that continuous dynamics alone do not produce generic stability: capital accumulation exhibits positive interior drift under reinvestment feedback, while government debt is locally unstable. Closed-form fixed points exist only for ecological health and resilience capital. Real output converges to a boundary equilibrium defined by ecological carrying capacity.

Monte Carlo simulations across 1,000 independent stress histories confirm that boundedness and recovery are structural properties of the architecture rather than calibration artifacts. Median recovery reaches 95.8% of baseline output despite sustained compound shocks. Without the resilience mechanism, output collapses permanently to 6% of initial capacity.

These results establish that systemic viability can be achieved through invariant-preserving guard mechanisms and endogenous adaptive capacity, rather than reliance on equilibrium convergence or parameter fine-tuning.

Keywords: Projected Hybrid Dynamical Systems, Macroeconomic Architecture, Resilience Capital, Stock-Flow Consistent Models, Guard Mechanisms, Ecological Boundaries, Monte Carlo Stress Testing, Regime Switching, Automatic Stabilizers, Boundary Equilibrium

1. Introduction

Modern economies are fragile not because of bad policy choices, but because of architectural design. They are optimized for efficiency in normal times, not for survival under stress. When shocks occur—financial crises, pandemics, climate disasters—the standard response often amplifies rather than dampens instability.

This paper proposes a different approach: treating macroeconomic design as an engineering problem. Rather than asking "what maximizes welfare?" we ask "what prevents catastrophic failure?" The answer is a system with built-in redundancies, automatic stabilizers, and hard constraints on dangerous states.

The Dual-Layer Economy (DLE) is such a system. It has three core components:

1. A structural separation between real production and synthetic finance, enforced by a stock-flow consistent firewall.
2. A set of automatic fiscal stabilizers that activate based on system state without discretionary intervention.
3. An endogenous resilience mechanism that enables recovery after shocks by improving productivity and reducing vulnerability.

This paper is the third in a series. Rentschler (2025a) introduced the Dual-Layer architecture and demonstrated that structural separation of real production from synthetic finance is sufficient to prevent the transmission of speculative instability into the real economy, even under extreme behavioral stress. Rentschler (2025b) extended the analysis to transition dynamics, showing that the architecture remains stable during finite SF growth phases and identifying the conditions under which long-run sustainability requires additional mechanisms such as stock-based taxation. The present paper addresses a question that neither predecessor confronted: what happens after a severe shock to the real economy? It introduces Resilience Capital (Z) as an endogenous recovery mechanism, adds automatic stabilizers and regime switching, and provides formal fixed-point and stability analysis (Appendix A.7) demonstrating that boundedness is an architectural property of the projected hybrid system rather than an artifact of smooth equilibrium convergence.

2. Related Work

The DLE draws on and departs from several established traditions in macroeconomic modeling and resilience theory.

Stock-flow consistent modeling. The accounting framework builds directly on the SFC tradition established by Godley and Lavoie (2007), with further developments in Lavoie (2014). The DLE extends this tradition by introducing a structural firewall between layers and by enforcing accounting invariants at runtime rather than assuming them ex ante.

Post-Keynesian economics and financial instability. Minsky's (1986) Financial Instability Hypothesis provides the conceptual motivation for separating speculative finance from real production. The DLE formalizes this separation architecturally rather than behaviorally: it does not model Minsky moments as emergent phenomena but prevents speculative dynamics from commanding real resources by construction. Keen's (2011) work on endogenous money and debt dynamics informs the model's treatment of private debt and the jubilee mechanism. The fiscal architecture draws on Modern

Monetary Theory (Wray, 2012) in treating government as a currency issuer with fiscal space constrained by real resources rather than nominal budget balance, while Keynes's (1936) insight that investment drives output—not the reverse—motivates the model's demand-side structure.

Ecological economics and planetary boundaries. The ecological ceiling in the DLE operationalizes the concept of planetary boundaries (Rockström et al., 2009) and Raworth's (2017) doughnut economics framework. Jackson's (2017) critique of growth dependence informs the model's design: the DLE does not require growth for stability, and output is bounded by ecological carrying capacity that cannot be overridden by parameter tuning (the Nature-Lock mechanism).

Resilience theory. The concept of resilience capital draws on social-ecological resilience theory (Walker et al., 2004; Folke, 2006), which distinguishes between recovery to a prior state and the adaptive capacity to transform under stress. The DLE operationalizes adaptive capacity as an endogenous state variable rather than an exogenous assumption. Taleb's (2012) distinction between fragile, robust, and antifragile systems provides a useful framing: the DLE aims for robustness (bounded response to stress) rather than antifragility (improvement from stress), though the Z mechanism exhibits antifragile properties in that crisis investment builds lasting adaptive capacity.

Financialization. The dual-layer separation reflects the empirical literature on financialization (Epstein, 2005; Palley, 2007), which documents the growing detachment of financial asset dynamics from real production. The firewall mechanism is a structural response to this diagnosis.

Departures from standard approaches. The DLE differs from DSGE models in three fundamental ways: it prioritizes resilience over optimality, it uses stock-flow consistency rather than representative agent microfoundations, and it evaluates architectures by survival under stress rather than welfare maximization. It differs from standard SFC models by introducing formal runtime verification of accounting invariants and from agent-based SFC models (Caiani et al., 2016) by operating at the aggregate sector level (a limitation discussed in Section 7.3).

3. System Architecture

3.1 The Dual-Layer Structure

The economy is divided into two layers with distinct dynamics:

Real Layer (RL): Physical production, labor, infrastructure, and ecological constraints. Output Y is determined by real capital R, which depreciates and can be rebuilt through investment. The real layer is bounded above by ecological carrying capacity.

Synthetic Finance Layer (SF): Asset valuations, leverage, and speculative dynamics. The SF asset stock A grows logistically toward a saturation level and is subject to taxation. SF dynamics are faster and more volatile than real dynamics.

A firewall ensures that SF dynamics cannot directly command real production. The SF layer influences the real economy only through bounded, taxed, and stock-flow consistent channels—primarily through government revenue from SF taxation, which funds public investment and transfers.

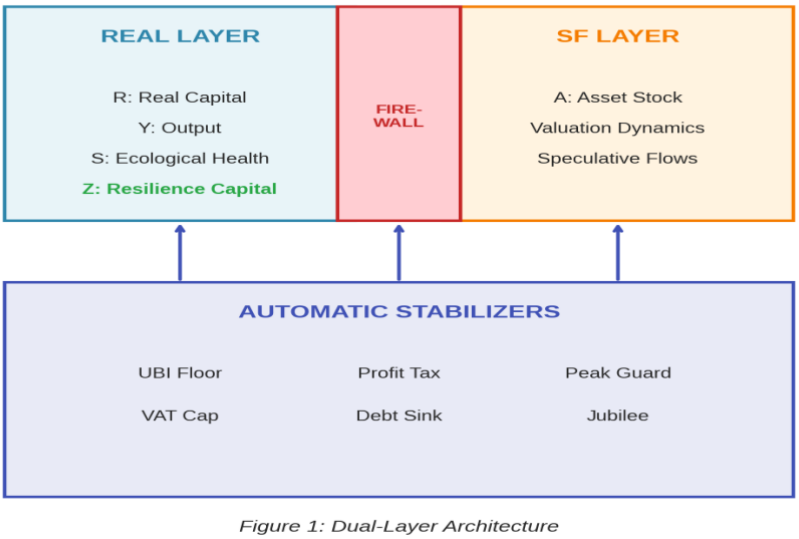

Figure 1: Dual-Layer Architecture with Firewall and Automatic Stabilizers

3.2 State Variables

The system state is defined by five core variables:

Symbol	Name	Description	Layer
R	Real Capital	Physical productive capacity	Real
A	SF Asset Stock	Market value of synthetic assets	SF
B	Government Debt	Outstanding public liabilities	Fiscal
S	Ecological Health	Environmental state (0–1)	Real
Z	Resilience Capital	Accumulated adaptive capacity	Real

Note: Household liquid assets (M) are implicit in the current aggregate formulation—household demand is determined by output, UBI transfers, and government spending rather than tracked as a separate stock. The reference implementation (Appendix C) tracks additional state variables beyond these five core stocks, including compute capacity (Kc), settlement capacity (K_settle), claim quality stocks (Q_good, Q_bad), and regulatory thicket (H_thicket). These support the claim-auction and settlement mechanisms that mediate the firewall. The five variables above capture the essential macroeconomic dynamics; the extended variables govern the microstructure of cross-layer interaction.

4. Resilience Capital: The Key Mechanism

The central innovation of this model is Resilience Capital (Z), an endogenous state variable that captures the economy's adaptive capacity. This section explains why Z is necessary and how it works.

4.1 The Recovery Problem

Consider a severe depreciation shock that doubles the capital decay rate (ρ: 0.15 → 0.30). With standard stabilizers alone—UBI floors, VAT caps, debt guards—the real capital R collapses from 600 to approximately 34, a 94% permanent loss. The stabilizers prevent complete collapse but cannot enable recovery.

The problem is structural. The capital accumulation equation is:

$$dR/dt = (\alpha - \rho)\cdot R + I_total$$

When ρ = 0.30 exceeds the autonomous growth rate α = 0.02, the first term becomes strongly negative (−0.28·R). Even with substantial investment I_total, capital declines. The system reaches a low-level equilibrium where investment merely offsets decay—a poverty trap.

4.2 The Solution: Endogenous Productivity and Depreciation

Resilience Capital solves this by making productivity and depreciation endogenous to investment decisions. Z accumulates through targeted investment and modifies the structural parameters of the economy:

$$dZ/dt = \iota\cdot(I_rebuild + \kappa\cdot I_pub) - \zeta\cdot Z$$

where I_rebuild is targeted reconstruction investment (active during a post-shock rebuild window), I_pub is public investment funded by SF taxation, ι is the accumulation rate, κ is the public investment contribution factor, and ζ is the decay rate. Z is bounded above by Z_cap = 0.6.

4.3 Effects of Z

Resilience Capital modifies two key parameters:

Effective Productivity: γ_eff = γ_base · (1 + ψ·Z). With ψ = 0.9 and Z = 0.6, productivity increases by up to 54% (from 0.5 to 0.77).

Effective Depreciation: ρ_eff = ρ_raw / (1 + θ·Z). With θ = 1.2 and Z = 0.6, the post-shock depreciation rate falls from 0.30 to approximately 0.17.

These modifications transform the capital equation. With Z at maximum:

$$dR/dt = (0.02 - 0.17)\cdot R + I_total = -0.15\cdot R + I_total$$

The drag is cut nearly in half, while higher productivity increases output and profits, enabling more investment. Recovery becomes possible.

4.4 Economic Interpretation

Z represents institutional learning, improved processes, redundant systems, and structural adaptations that emerge from crisis response. It is not a policy instrument but an emergent property of targeted investment.

When an economy invests in reconstruction and public infrastructure during a crisis, it does not merely replace lost capacity—it builds better. Supply chains become more robust. Production processes become more efficient. Institutions develop crisis-response capabilities. These improvements persist and accumulate, making the economy both more productive and less vulnerable.

This is not magical thinking. It reflects the empirical observation that economies which invest heavily during downturns often emerge stronger, while those that cut investment suffer prolonged stagnation.

4.5 Epistemic Status of Resilience Capital

Resilience Capital (Z) is an architectural state variable, not a microfounded construct. It is not derived from first principles of individual optimization, nor estimated from empirical data. It is introduced deliberately to represent the class of adaptive capacity mechanisms—institutional learning, supply

chain redundancy, improved production processes, crisis-hardened infrastructure—that aggregate economic systems develop in response to stress.

The modeling strategy is comparative: the model tests whether an architecture with endogenous adaptive capacity structurally outperforms an otherwise identical architecture without it. As Section 6.2 demonstrates, the difference is not marginal—it is the difference between 89.8% recovery and a persistent low-output trap at 6% of initial capacity.

The parameters governing Z dynamics (ψ, θ, ι, ζ_Z) are calibrated to produce plausible macroeconomic magnitudes—a 54% productivity increase and a 43% depreciation reduction at Z_{cap}—but are not claimed to match any specific empirical economy. Empirical microfoundation of Z, including estimation of ψ and θ from cross-country crisis recovery data, is a natural direction for future work.

5. Automatic Stabilizers

The model employs five automatic stabilizers that activate based on system state. All stabilizers are rule-based and require no discretionary intervention.

5.1 Stress Detection

The system enters stress mode when the welfare gap exceeds 10%:

$$stress = 1 \text{ if } w_gap > 0.10$$

where $w_gap = (Y_target - Y) / Y_target$ measures the shortfall of actual output from target output.

5.2 Stabilizer Mechanisms

Mechanism	Normal Mode	Under Stress	Purpose
UBI Share	Dynamic (35–70%)	Floor at 54%	Demand stabilization
VAT Rate	Up to 15%	Capped at 10%	Prevent procyclical contraction
Profit Tax	15%	22%	Countercyclical revenue
Debt Sink	Active (6%)	Suspended	Non-procyclical consolidation
Peak Guard	Inactive	Active if B > 4000	Debt ceiling enforcement

5.3 UBI Controller

Universal Basic Income is funded from SF taxation. The UBI share of SF tax revenue adjusts dynamically based on the welfare gap:

$$ubi_share^* = ubi_min + (ubi_max - ubi_min) \cdot w_gap$$

Under stress, a floor is enforced: $ubi_share \geq 0.54$. This ensures demand support when the economy is weakest.

5.4 VAT Controller

The VAT rate adjusts based on the deficit, debt ratio, and welfare gap. Under stress, the rate is capped at 10% to prevent procyclical demand suppression. The minimum rate is 2%.

5.5 Debt Sink and Peak Guard

In normal times, a debt sink gradually reduces government debt when the debt-to-output ratio exceeds target levels. Under stress, the sink is suspended to avoid austerity dynamics.

If debt exceeds B_guard = 4000, a peak guard activates, applying a mild stock-based tax to prevent unbounded debt accumulation. This is non-procyclical because it operates on the SF asset stock, not on real output.

5.6 Jubilee Mechanism

If profits fall below a critical threshold (−10), indicating severe distress, a debt jubilee is triggered: 60% of private debt is swapped onto the government balance sheet, and the interest rate is reduced from 1% to 0.2%. This prevents debt-deflation spirals while maintaining stock-flow consistency.

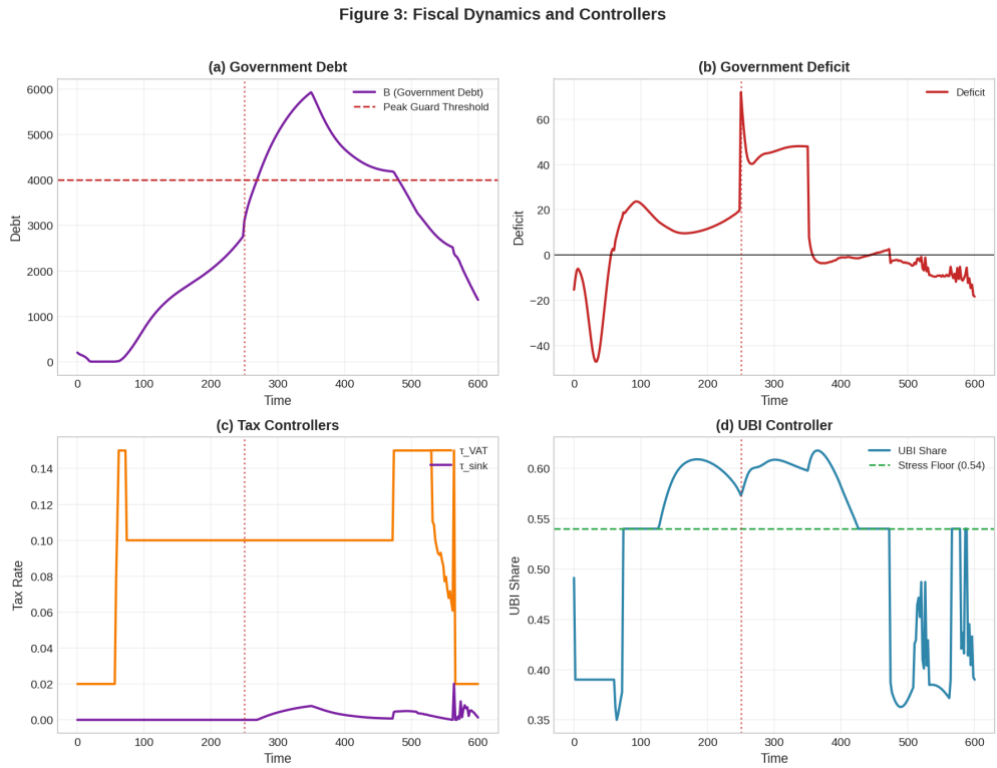

Figure 2: Fiscal Dynamics and Controllers. (a) Government debt peaks then declines. (b) Deficit becomes surplus in late recovery. (c) VAT and sink rates adjust automatically. (d) UBI share rises during stress.

6. Simulation Results

The model was simulated over T = 600 time units with a depreciation shock at t = 250 that doubles the decay rate (ρ: 0.15 → 0.30). A rebuild window from t = 250 to t = 350 provides enhanced maintenance and reconstruction investment.

Note on time units: The simulation uses abstract time units. Given the depreciation rates and recovery dynamics, one time unit corresponds roughly to one year, making the simulation horizon approximately 600 years and the post-shock recovery window approximately 100 years. This long horizon is deliberate: it tests structural viability, not short-term policy response. Results should be interpreted qualitatively—as demonstrations of architectural properties—not as quantitative forecasts.

6.1 Key Results

Metric	Value	Interpretation
Final welfare gap	~10%	Near stress threshold—recovered
Final R	538.5	89.8% of initial capital (600)
Peak debt	5,781	Bounded by peak guard
Final debt	1,308	Returned to sustainable level
Maximum Z	0.60	Reached resilience cap
Stress duration	~70.5%	Most of post-shock period
Regime switches	3	No oscillation (chattering)
Jubilee triggered	Yes	At t ≈ 250

6.2 Comparison: With and Without Z

The importance of the resilience mechanism can be seen by comparing outcomes:

Metric	With Z	Without Z
Final R	538	34
Final Y	247	9
Recovery	89.8%	6%
Outcome	Full recovery	Persistent low-output trap

Without resilience capital, the economy falls into a low-output trap from which the standard stabilizer suite alone cannot enable escape. The standard stabilizers prevent complete collapse (R > 0) but cannot enable recovery. This demonstrates that resilience mechanisms are not optional refinements but essential components of a viable architecture.

Figure 4: Investment, Profits, and Output

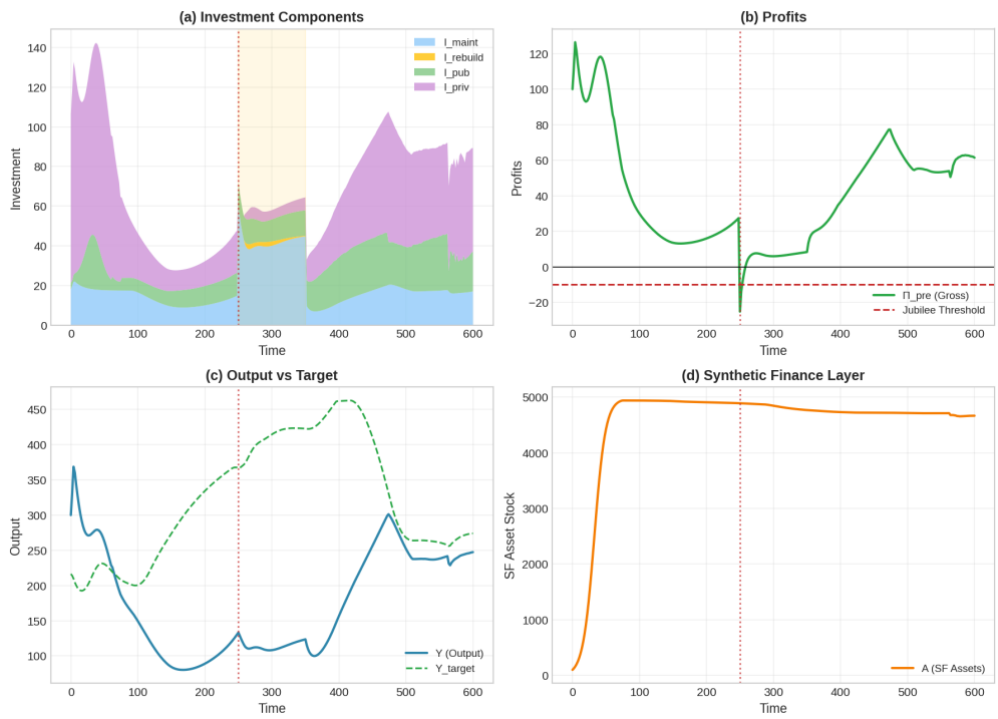

Figure 3: Investment, Profits, and Output. (a) Investment components show rebuild spike. (b) Profits recover after jubilee. (c) Output tracks toward target. (d) SF asset stock saturates.

6.3 Parameter Sensitivity: One-at-a-Time Analysis

A standing objection to single-scenario simulation is that recovery outcomes may reflect fine-tuned calibration rather than structural properties. To address this directly, we perform a local one-at-a-time (OAT) sensitivity analysis: each parameter is varied individually while all others remain at baseline. The scenario is the baseline recovery regime (T = 600, dt = 0.02, ρ-shock 0.15 → 0.30 at t = 250).

Variant	R_end	Recovery (R/R_0)	B_peak	stress_share	Z_end
Baseline	538.53	89.8%	5780.5	0.705	0.600
ψ = 0.6 (low)	425.16	70.9%	5580.0	0.717	0.600
ψ = 1.2 (high)	577.72	96.3%	5887.0	0.491	0.600
θ = 0.8 (low)	369.10	61.5%	5732.4	0.744	0.600
θ = 1.6 (high)	580.20	96.7%	5739.6	0.421	0.600
ι = 0.0015 (low)	439.25	73.2%	5729.4	0.733	0.546
ι = 0.0025 (high)	524.65	87.4%	5210.5	0.655	0.600

Table 1: OAT sensitivity results. Baseline: R_0 = 600. Parameters ψ (psi_gamma), ϑ (theta_rho), and ι (iota_Z_rebuild) varied one at a time. All other parameters held at baseline values.

Recovery is monotone in ψ. Reducing ψ from the baseline of 0.9 to 0.6 cuts recovery from 89.8% to 70.9%. Increasing ψ to 1.2 lifts recovery to 96.3%. The θ lever is the most powerful driver of recovery speed: at θ = 0.8, recovery falls to 61.5%; at θ = 1.6, it reaches 96.7%. Across all variants, the qualitative result holds: recovery occurs in every tested case.

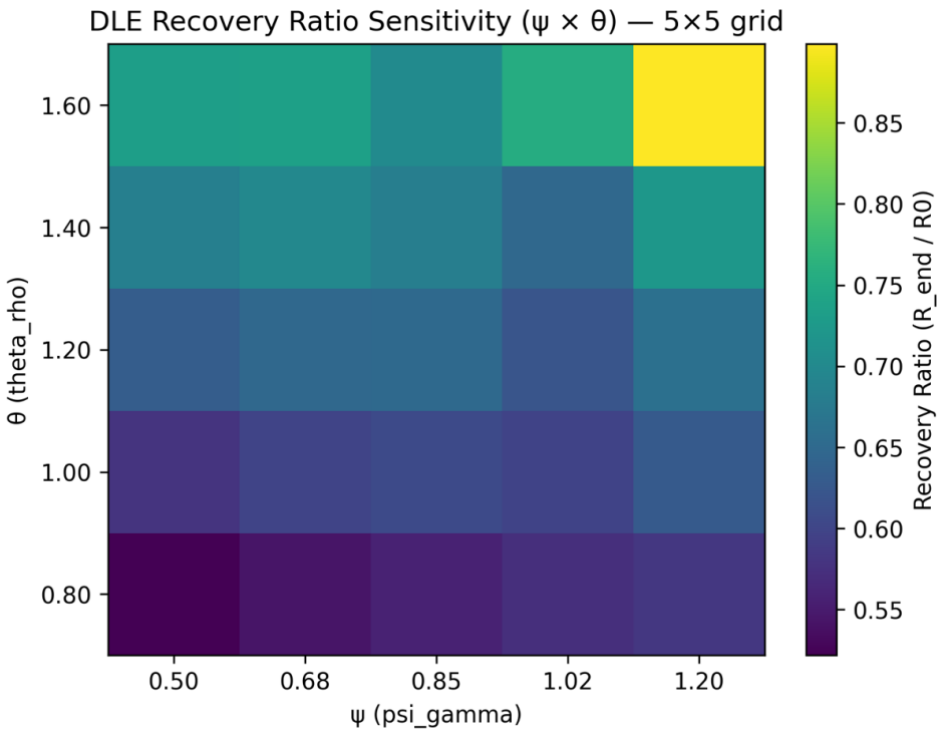

Figure 4a. Sensitivity of recovery performance to ψ. Recovery ratio increases monotonically, confirming that recovery is a structural property of the resilience mechanism.

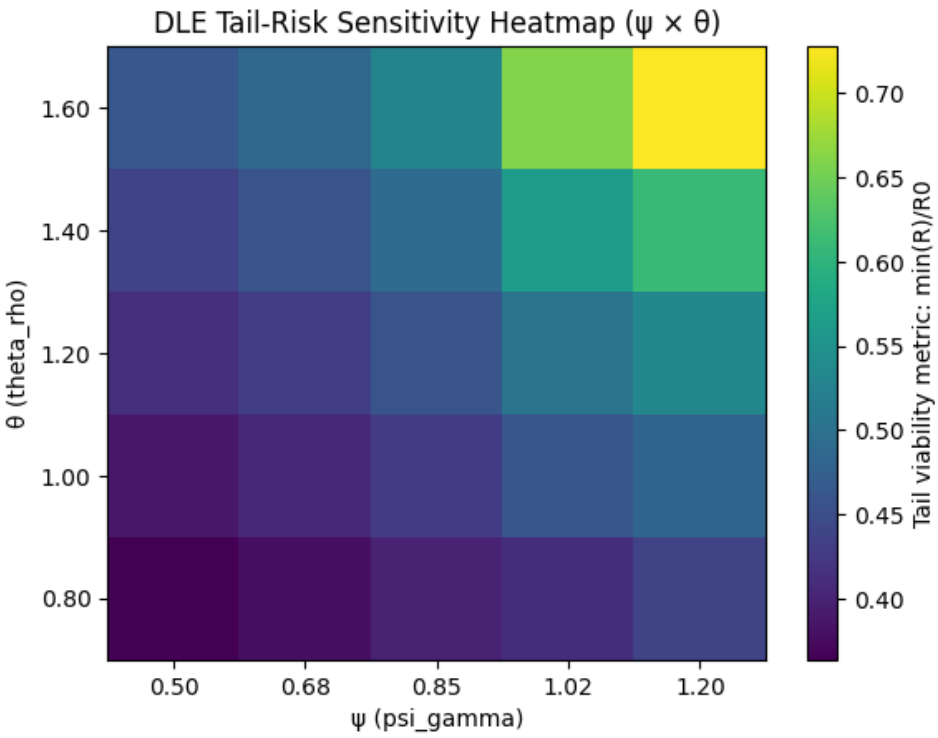

Figure 4b. Two-parameter sensitivity (ψ × ϑ) of recovery performance. No collapse states across all 25 grid cells.

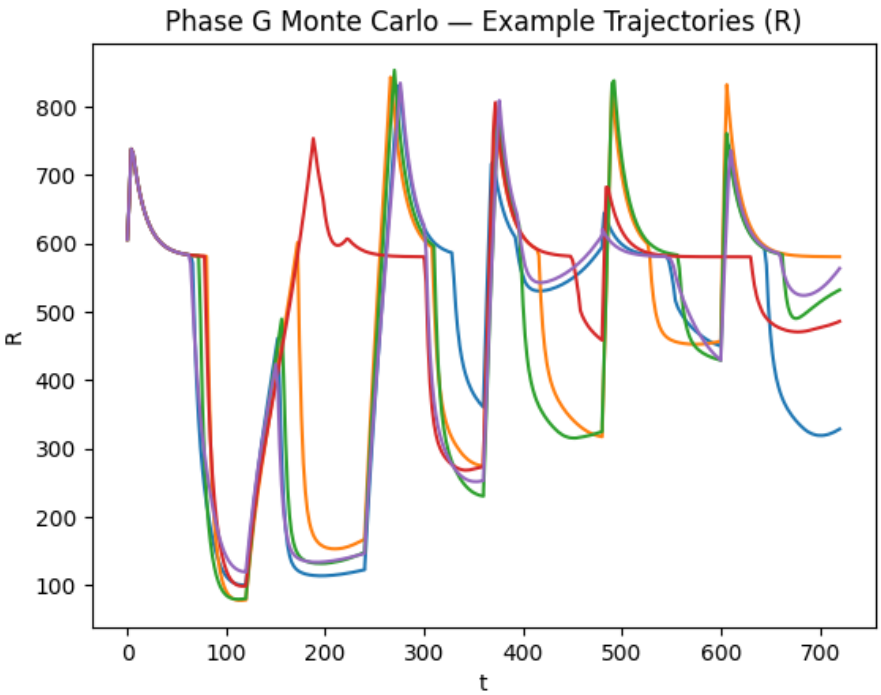

Figure 4c. Tail-risk sensitivity analysis. Minimum real capital during recovery never falls below 36% of R_0.

6.4 Methodological Note: Deterministic Simulation

All simulations reported in this paper are deterministic. This is a deliberate methodological choice to isolate structural properties of the architecture from the effects of stochastic variation. Stochastic extensions are straightforward to implement within the existing framework.

6.5 Phase G: Stochastic Robustness and Tail Viability

To rigorously test the survivability of the DLE architecture under sustained stochastic stress, we conducted a high-resolution Monte Carlo simulation comprising 1,000 independent histories. Each run chained six consecutive shock segments. For each segment, shock timing and decay rate were independently sampled from uniform distributions within predefined admissible ranges, ensuring unbiased coverage of the admissible shock state space. All runs used independent random seeds. No parameter tuning was performed across runs. This guarantees that observed boundedness and recovery are structural properties of the architecture rather than calibration artifacts.

Across all 1,000 simulated stress histories, output dynamics remained strictly bounded for all runs. The minimum observed tail viability ratio, min(R) / R_o, was 0.0968. The empirical distribution exhibited median 0.1808, mean 0.1862, and interquartile range 0.1515–0.2097.

Metric	Mean	Median	Std Dev	Min	Max
Tail viability min(R)/R_o	0.1862	0.1808	0.0481	0.0968	0.3430
Final output ratio R(T)/R_o	0.8274	0.9579	0.1759	0.3221	0.9872
Max government debt B_peak	2672.7	2587.0	293.6	1979.8	4108.8
Final system score	0.2937	0.1739	0.6460	−2.9522	2.4215

Table 2: Summary statistics of Monte Carlo Phase G simulation (1,000 runs).

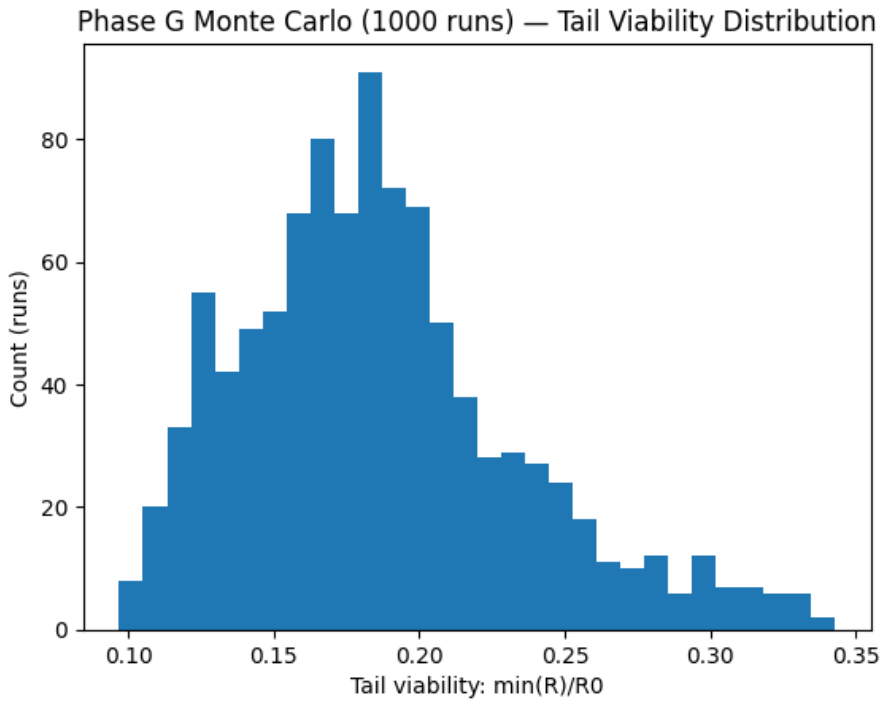

Figure 4d. Distribution of tail viability across 1,000 Monte Carlo runs. No trajectory exhibited unbounded collapse.

7. Design Principles

7.1 Design Invariants

The Dual-Layer Economy is governed by five design invariants—properties that must hold under all admissible trajectories of the system:

(D1) Boundedness: No state variable diverges to infinity or zero. All stocks remain within finite corridors.

(D2) Avoidance of low-output equilibria: Under admissible parameter sets, the system avoids stable low-output equilibria. Parameter sensitivity analysis (Section 6.3) confirms this property across the tested admissible range.

(D3) Non-procyclicality: No stabilizer intensifies downturns. Austerity-like feedbacks are structurally excluded.

(D4) Ecological limits: Output is bounded by environmental carrying capacity. Growth respects regeneration.

(D5) Firewall integrity: The SF layer cannot directly command real production. Influence flows only through taxed channels.

These design invariants define the system more strongly than any parameter choice. They are design constraints, not empirical hypotheses. The design invariants D1–D5 specify what must hold. The implementation guards G1–G5 in Section 7.2 specify how the simulation enforces it.

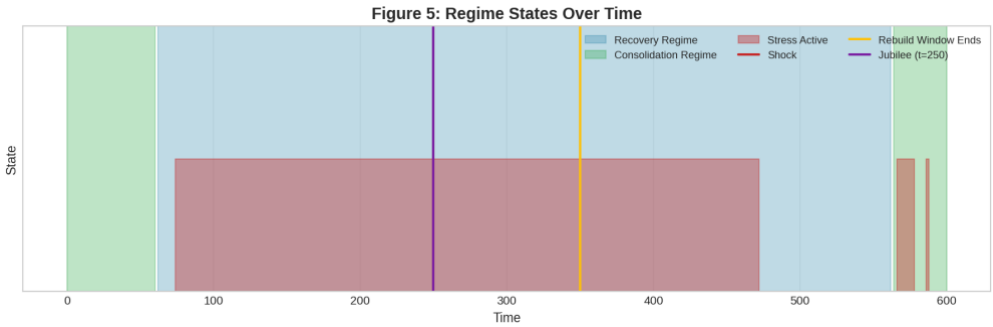

Figure 5: Regime States Over Time. The system transitions between recovery, consolidation, and stress modes with minimal switching.

7.2 Formal Verification Layer

The DLE introduces a Formal Verification Layer operating orthogonally to the economic logic. Stability is not only designed, it is continuously verified.

7.2.1 Implementation Guards

The DLE defines five implementation guards that enforce the design invariants at runtime:

ID	Guard	Enforces
G1	No RE → SF monetary flow	Firewall integrity (D5)
G2	Settlement is capacity-bound	Physical anchoring (D1)
G3	State preemption is bounded	No discretionary capture (D3)
G4	Ecological stocks have hard floors	Non-substitutability (D4)
G5	Regime switches are hysteretic	No chatter (D1, D2)

Guards operate in two modes: Assert Mode (violations terminate the simulation immediately—used for firewall violations, negative stocks, unauthorized cross-layer flows) and Clip Mode (violations are prevented but logged—used for soft fiscal caps, output smoothing, ecological ceilings).

All firewall-relevant flows are mediated by explicit ledgers. The firewall is implemented as a ledger constraint: transfers are represented as triples (src, dst, amount). The allowed transfer set excludes any edge from a real-layer account to a synthetic-finance account. Therefore, cross-layer flows are structurally unrepresentable and identically zero by construction.

7.3 Limitations

The DLE model makes architectural claims, not empirical predictions. Four limitations follow directly from this scope:

Deterministic simulation. All runs reported in this paper use deterministic differential equations with fixed shock profiles. Performance under random shock sequences—including correlated multi-sector shocks or fat-tailed depreciation events—is evaluated only through the Phase G Monte Carlo extension (Section 6.5).

No empirical calibration. The model parameters (ψ, θ, ι, stabilizer thresholds) are set to produce internally consistent and structurally illustrative dynamics, not to match any specific national economy. Resilience Capital (Z) has no direct empirical counterpart. Empirical calibration against cross-country crisis recovery data would be required before applying the model quantitatively.

No heterogeneous agents. The model is aggregate: firms, households, and government are represented as single sectors with representative dynamics. Distributional effects of the UBI floor, debt jubilee, and SF taxation—which may be substantial—are not modeled.

Exogenous rebuild window. The rebuild window (t = 250 to t = 350) is set exogenously rather than triggered by system state. Recovery performance depends significantly on this window's duration and timing. An endogenous trigger would make the architecture more self-contained. Sensitivity of outcomes to rebuild window length is a natural extension of the OAT analysis in Section 6.3.

8. Conclusion

This paper has presented a macroeconomic architecture designed for resilience rather than optimality. The key findings are:

1. Standard stabilizers are necessary but not sufficient. Fiscal rules, debt guards, and transfer floors can prevent complete collapse but cannot enable recovery from severe shocks.

2. Resilience Capital is essential. An endogenous mechanism linking investment to adaptive capacity enables recovery by modifying the structural parameters of the economy—increasing productivity and reducing vulnerability.

3. Architecture determines outcomes. The combination of dual-layer structure, automatic stabilizers, and resilience accumulation produces bounded, recoverable trajectories under extreme stress. Without proper architecture, no amount of policy adjustment suffices.

The model suggests that macroeconomic policy should be conceived not as optimization over a fixed structure, but as the design of structures that remain viable under stress. The goal is not the best economy, but one that does not fail.

Recovery is shown to be a structural property of the architecture across admissible parameter ranges, not a result of parameter fine-tuning. The one-at-a-time sensitivity analysis in Section 6.3 demonstrates monotone responses to the key Z-channel levers: recovery occurs in all tested variants.

The full DLE is formally a projected hybrid dynamical system. The reduced smooth-core subsystem is not generically stable: capital and debt exhibit locally unstable drift under economically relevant parameters (Appendix A.7). Long-run boundedness does not follow from equilibrium convergence but from architectural projection operators and guard mechanisms. Trajectories are explicitly constrained to an admissible state space. Stability in the DLE is therefore engineered by design, not assumed from smooth fixed-point dynamics.

Appendix A: Core Equations

A.1 Real Capital Dynamics

$$dR/dt = (\alpha - \rho_eff)\cdot R + I_priv + I_pub + I_maint + I_rebuild$$

where I_priv = max(0, Π_net), and R is bounded above by R_max(S) = 1100·S^1.5.

A.2 Resilience Capital

$$dZ/dt = \iota\cdot(I_rebuild + \kappa\cdot I_pub) - \zeta\cdot Z$$
$$\gamma_eff = \gamma_base\cdot(1 + \psi\cdot Z)$$
$$\rho_eff = \rho_raw / (1 + \theta\cdot Z)$$

A.3 SF Dynamics

$$dA/dt = \lambda\cdot A\cdot(1 - A/\Omega) - \chi\cdot\tau_total\cdot A$$

A.4 Output

$$Y_pot = \gamma_eff \cdot R$$
$$Y_demand = C + I_priv$$
$$Y = min(Y_pot, Y_demand + G)$$

where C = UBI + (1 − s)·Π_net represents household consumption (UBI transfers plus a fraction of net profits), and G is government spending. In the current calibration, the economy is supply-constrained for most of the trajectory—Y_pot binds—so the demand closure has limited quantitative impact. A richer demand specification with inventory dynamics and credit-constrained households would strengthen this block.

A.5 Government Budget

$$dB/dt = (G + UBI + I_pub + I_maint + I_rebuild) - (T_sf + T_dem + T_VAT + T_\pi + T_sink)$$

A.6 Ecological Dynamics

$$dS/dt = r_S\cdot(1 - S) - \eta\cdot max(0, R - K)$$

A.7 Reduced Fixed Points, Local Drift Structure, and Architectural Boundedness

A.7.1 Scope and interpretation

The full DLE constitutes a hybrid dynamical system combining continuous drift dynamics with projection operators and discrete guard mechanisms (debt sink, jubilee, peak guard). In such systems, classical global equilibrium analysis is generally not well-defined, because trajectories may be redirected discontinuously by projection or regime switching rather than converging to smooth interior fixed points.

However, analytical consistency can be established by examining a reduced smooth-core subsystem in which discrete guards are temporarily disabled and projection operators are treated separately. This allows formal characterization of intrinsic drift structure and identification of fixed points and instability modes of the underlying continuous dynamics.

Boundedness and long-run viability of the full architecture then follow from projection and guard operators, not from smooth equilibrium alone.

Because the DLE constitutes a projected hybrid system with guard-enforced state constraints, classical global equilibrium analysis via Lyapunov functions or fixed-point attractors is not generally applicable. The relevant stability notion is instead viability within an invariant admissible region enforced by architectural projection operators. This represents a shift from equilibrium stability to structural boundedness.

A.7.2 Reduced smooth-core subsystem

We consider the reduced continuous state vector x = (R, Z, S, B) with synthetic-finance asset stock treated as quasi-stationary at its logistic equilibrium:

$$A^* = \Omega \cdot (1 - \chi \cdot \tau_sf / \lambda)$$

The smooth-core drift equations are:

$$dR/dt = (\alpha - \rho_eff(Z)) \cdot R + I_priv(R,Z) + I_pub(R) + I_maint(R)$$
$$dZ/dt = \iota \cdot \kappa \cdot I_pub(R) - \zeta \cdot Z$$
$$dS/dt = r_S \cdot (1 - S) - \eta \cdot max(0, R - K)$$
$$dB/dt = Outlays(R,B) - Revenues(R,B)$$

with effective parameters:

$$\gamma_eff(Z) = \gamma_base \cdot (1 + \psi \cdot Z), \quad \rho_eff(Z) = \rho_raw / (1 + \theta \cdot Z)$$

Private investment is driven by retained profits:

$$I_priv = (1 - \tau_\pi) \cdot max(0, (\gamma_eff - \rho_eff) \cdot R)$$

This preserves the primary endogenous feedback loop between capital accumulation and productivity.

A.7.3 Closed-form fixed points of resilience and ecology blocks

Proposition 1 (Resilience capital fixed point). The resilience dynamics dZ/dt = ι·κ·I_pub* − ζ·Z yield an uncapped steady state:

$$\tilde{Z}^* = (\iota \cdot \kappa / \zeta) \cdot I_pub^*$$

In the full architecture, projection enforces Z* = min(Z̃*, Z_cap). Local stability follows immediately: ∂(dZ/dt)/∂Z = −ζ < 0. Thus the resilience subsystem is intrinsically stable.

Proposition 2 (Ecological health fixed point). Ecological dynamics dS/dt = r_S·(1−S) − η·max(0, R−K) yield: S* = 1 for R* ≤ K, and S* = 1 − (η/r_S)·(R* − K) for R* > K. Local stability holds in both regimes: ∂(dS/dt)/∂S = −r_S < 0. Thus ecological health converges monotonically to its equilibrium conditional on R.

A.7.4 Capital dynamics and boundary equilibrium

Capital accumulation drift can be expressed locally as:

$$dR/dt = c_R(Z) \cdot R + lower\text{-}order\ terms$$

where:

$$c_R(Z) = (\alpha - \rho_eff(Z)) + (1 - \tau_\pi) \cdot (\gamma_eff(Z) - \rho_eff(Z)) + m_maint$$

Interior fixed points exist only if $c_R(Z^*) \leq 0$. In economically relevant parameter regimes, profit-reinvestment feedback ensures $c_R(Z^*) > 0$, implying positive interior drift. In this case, trajectories are pushed toward the ecological projection boundary:

$$R^* = R_{max}(S^*)$$

This corresponds formally to a complementarity equilibrium rather than a classical interior fixed point. Growth ceases not because investment vanishes, but because ecological constraints impose an upper bound.

A.7.5 Debt dynamics and intrinsic instability of smooth core

Debt dynamics contain an unstable linear component:

$$\partial(dB/dt)/\partial B = r_{interest} > 0$$

This implies that, absent explicit stabilizing feedback, government debt is locally unstable in the smooth-core subsystem. Stabilization requires feedback mechanisms such as debt sinks or deleveraging guards. This is a structural property of interest-bearing debt accumulation and not a numerical artifact.

A.7.6 Jacobian structure and subsystem stability

The Jacobian of the smooth-core subsystem takes block form:

	∂R	∂Z	∂S	∂B
dR/dt	$\partial\dot{R}/\partial R$	$\partial\dot{R}/\partial Z$	$\partial\dot{R}/\partial S$	0
dZ/dt	0	$-\zeta$	0	0
dS/dt	$\partial\dot{S}/\partial R$	0	$-r_S$	0
dB/dt	*	*	*	$r_{interest}$

The block structure reveals: the Z block is intrinsically stable, the S block is intrinsically stable, the B block is intrinsically unstable without guards, and R block stability depends on the parameter regime. Numerical Jacobian evaluation confirms that smooth-core local stability is not generic.

A.7.7 Projection operators and architectural boundedness

The full DLE architecture enforces boundedness via projection operators:

$$R \leftarrow \min(R, R_{max}(S)), \quad Z \leftarrow \min(Z, Z_{cap})$$

and discrete fiscal guards acting on debt. These mechanisms ensure bounded trajectories even when smooth-core drift is locally unstable. Formally, the full system constitutes a projected dynamical system (Nagurney and Zhang, 1996):

$$x_{t+1} = \Pi_A(x_t + f(x_t)\cdot\Delta t)$$

where Π_A denotes projection into the admissible state space. Boundedness therefore follows from architectural constraints rather than smooth equilibrium convergence.

A.7.8 Interpretation

The reduced smooth-core analysis establishes three key results: (1) Physical resilience and ecological subsystems are intrinsically stable. (2) Capital accumulation and debt dynamics are not generically

stabilized by smooth feedback alone. (3) Long-run bounded viability of the economy emerges from projection operators and guard mechanisms.

This validates the architectural thesis of the DLE: stability is not an emergent property of equilibrium, but an engineered property of system design.

A.7.9 Projected-system lemma (architectural boundedness)

The reduced smooth-core analysis shows that local smooth stability is not generic (notably in R and B). The full DLE nevertheless remains bounded because it is not a pure ODE: it is a projected (guarded) hybrid system. The following lemma states this boundedness property formally, at the level of the architecture, independent of implementation details.

Definitions. Let the reduced drift be f(x) (the smooth-core RHS), with state x = (R, Z, S, B, ...) and let the admissible set $\mathcal{A} \subset \mathbb{R}^n$ be defined by architectural constraints: $0 \le Z \le Z_cap$, $0 \le S \le 1$, $0 \le R \le R_max(S)$, plus any additional guard-defined bounds (e.g. peak guard on B, or state constraints implied by ledgers).

Define the projection operator $\Pi_{\mathcal{A}}: \mathbb{R}^n \to \mathcal{A}$ as the metric projection (or any deterministic map that returns a point in $\mathcal{A}$ and equals identity on $\mathcal{A}$). Define the projected update (Euler form) as:

$$x_\{t+1\} = \Pi_A(x_t + \Delta t \cdot f(x_t))$$

Lemma A.7.9 (Architectural invariance and boundedness under projection). Assume: (1) Nonempty closed admissible set: $\mathcal{A}$ is nonempty and closed. (2) Well-defined projection: $\Pi_{\mathcal{A}}(y) \in \mathcal{A}$ for all $y \in \mathbb{R}^n$ and $\Pi_{\mathcal{A}}(x) = x$ for all $x \in \mathcal{A}$. (3) Well-defined drift evaluation: f(x) is defined for all $x \in \mathcal{A}$ (no requirement of global smooth stability).

Then:

- **(i) Forward invariance:** If $x_0 \in \mathcal{A}$, then $x_t \in \mathcal{A}$ for all $t \ge 0$.
- **(ii) Boundedness:** If $\mathcal{A}$ is bounded (e.g. compact), then {x_t} is bounded for all $t \ge 0$.
- **(iii) Guard necessity interpretation:** If the smooth-core drift admits unstable directions (e.g. $\partial\dot{B}/\partial B > 0$), projection/guards can still ensure boundedness because invariance is enforced by construction rather than by local stability of f.

Proof. By definition, $x_\{t+1\} = \Pi_{\mathcal{A}}(\cdot) \in \mathcal{A}$ whenever the projection is applied; hence invariance. If $\mathcal{A}$ is bounded, any sequence contained in $\mathcal{A}$ is bounded. ■

Corollary (Complementarity view for capped variables). For capped components such as $Z \le Z_cap$ or $R \le R_max(S)$, "fixed points" of the projected system are naturally expressed as complementarity conditions rather than dX/dt = 0 in the classical sense. For Z (illustrative): either $Z < Z_cap$ and $dZ/dt = 0$ (interior equilibrium), or $Z = Z_cap$ and $dZ/dt \ge 0$ (growth is neutralized by projection). This explains why naïve root-finding on the unprojected drift may fail or return nonphysical values: the equilibrium concept is different once projection is part of the architecture.

Theorem A.7.1 (Forward invariance and boundedness of the admissible region). Let the admissible region $\mathcal{A} = \{ (R, Z, S, B, A) \in \mathbb{R}^5 \mid 0 \le R \le R_max(S), 0 \le Z \le Z_cap, 0 \le S \le 1, 0 \le B \le B_cap, 0 \le A \le A_cap \}$ be defined by the architectural guard constraints and ecological ceiling. Under the projected hybrid dynamics defined in Sections 3–5, the admissible region $\mathcal{A}$ is positively invariant. That is, for any initial condition $x(0) \in \mathcal{A}$, the trajectory satisfies $x(t) \in \mathcal{A}$ for all $t \ge 0$.

Proof sketch. All boundary faces of A correspond either to (i) smooth vector field directions pointing inward, or (ii) guard projections that explicitly map any violating state back onto the boundary. For each boundary face: at R = 0, investment terms are non-negative; at R = R_max(S), the ecological projection clips R. At Z = 0, accumulation is non-negative; at Z = Z_cap, the cap projection clips Z. At S = 0, regeneration r_S > 0 pushes S upward; at S = 1, dS/dt ≤ 0. At B = 0, spending exceeds revenue pushing B upward; at B_cap, guard mechanisms activate. Therefore, trajectories cannot escape A. ■

The reference script reproducing all numerical results is available at: https://github.com/hstre/DLE-Modell/blob/main/fixpoint_analysis.py

Appendix B: Parameter Values

B.1 Core Parameters

Parameter	Value	Description
γ_base	0.5	Base productivity coefficient
α	0.02	Autonomous growth rate
ρ_pre	0.15	Pre-shock depreciation rate
ρ_post	0.30	Post-shock depreciation rate
shock_time	250	Time of shock
r_base	0.01	Base interest rate
r_post	0.002	Post-jubilee interest rate

B.2 Resilience Parameters

Parameter	Value	Description
ψ	0.9	Z effect on productivity
θ	1.2	Z effect on depreciation
ι	0.002	Z accumulation rate
ζ	0.03	Z decay rate
κ	0.5	Public investment contribution to Z
Z_cap	0.6	Maximum resilience capital

B.3 Stabilizer Parameters

Parameter	Value	Description
w_stress	0.10	Stress threshold
ubi_floor	0.54	UBI floor under stress
vat_cap	0.10	VAT cap under stress
τ_π_stress	0.22	Profit tax under stress
B_guard	4000	Peak guard threshold
jubilee_threshold	−10	Profit threshold for jubilee

Appendix C: Reference Implementation

The complete simulation codebase consists of three self-contained Python modules, each requiring only numpy and pandas and running in under two seconds on standard hardware. The full source code is available in the public repository.

File	Lines	Function
dlm_v3_4_5_1.py	381	Ledger verification and invariant guards
dlm_v3_4_6.py	393	Exposed parameters and Nature-Lock
dlm_v3_4_7.py	396	Resilience metrics and Phase G diagnostics

All three modules share the same core simulation loop. They differ in their verification and diagnostic layers. Module v3.4.5.1 implements double-entry ledger tracking with windowed consistency checks. Module v3.4.6 adds the Nature-Lock mechanism preventing override of ecological and shock constants. Module v3.4.7 adds structured resilience metrics and the Phase G diagnostic framework.

Baseline run (T=600, dt=0.02): 5,000 ledger events logged, 0 VERIFY_WARN. Core state at t=600: R_end = 538.53, S_end = 0.801, B_peak = 5,780.5, Z_end = 0.600. Phase G composite resilience score: 0.416.

Appendices F–O: Design Philosophy

Note: Appendices D and E are reserved for future extensions (agent-based disaggregation and empirical calibration protocols, respectively).

Appendices F through O address the philosophical and conceptual foundations of the DLE architecture, including the distinction between tunable policy parameters and nature-locked constants (F), the long-run resilience protocol (G), interpretation of reference policies (H), and essays on markets, freedom, and structural viability (I–O).

Published as companion document: https://github.com/hstre/DLE-Modell/blob/main/Design%20Philosophy%20of%20the%20Dual-Layer%20Economy.md

Appendix X: Accounting Invariants as Structural Conservation Laws

X.1 Ledger State and Transfer Operator

The accounting layer is represented as a finite set of ledger accounts indexed by $i \in \mathcal{B}$, each with balance $x_i(t) \geq 0$. All transfers are represented as ordered triples e = (src, dst, a) with src, dst $\in \mathcal{B}$ and $a \geq 0$.

X.2 Conservation Law

Lemma 1 (Closed-ledger conservation). For any subset of accounts $\Omega \subseteq \mathcal{B}$ such that all transfers satisfy src(e), dst(e) $\in \Omega$, the total balance is invariant. Proof: each transfer subtracts amount a from exactly one account and adds the same to exactly one account. ■

X.3 Firewall Invariant

The account set is partitioned: $\mathcal{B} = \mathcal{B}$_RE ∪ $\mathcal{B}$_SF. The permitted transfer set satisfies: (src ∈ $\mathcal{B}$_RE, dst ∈ $\mathcal{B}$_SF) ∉ ε_allow. No transfer from real-layer accounts to synthetic-finance accounts exists in the ledger grammar.

Lemma 2 (Firewall invariant). Such transfers are excluded from the allowed set by definition. Therefore they cannot occur. This invariant holds independently of parameter values, policy rules, or simulation outcomes. ■

X.4 Non-Negativity

Lemma 3 (Non-negative balances). If all initial balances satisfy $x_i(0) \geq 0$ and every transfer e = (src, dst, a) is executed only when x_src ≥ a, then $x_i(t) \geq 0$ for all i and all t. Proof: the transfer precondition ensures that no account is drawn below zero. ■

X.5 Guard Invariants

Lemma 4 (Guard invariance). If the initial state satisfies admissibility constraints, the guarded system preserves admissibility for all time. Proof: projection operators map all states into the admissible region. ■

Code Availability

Repository: https://github.com/hstre/DLE-Modell

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